

Date: _____

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Lab No. 8

→ Convert Infix → PostFix

Infix:

1. $A + B^*(C^D - E)^{(F + G^*H)} - I$

Post fix:

• $CD^E -$

• $FGH^* +$

• $(CD^E -)(FGH^* +)^$

• $B(CD^E - FGH^* +)^*$

• $AB(CD^E - FGH^* +)^* +$

• $AB(CD^E - FGH^* +)^* + I -$

→ $ABCD^E - FGH^* +^* + I -$

2. $A + (B * C - (D / E^F)^* G) + H$

Postfix:

• $EF^$

• $DEF^/$

• $DEF^/G^*$

• BC^*

• $BC^*DEF^/G^* -$

• $(BC^*DEF^/G^* -)H^*$

• $ABC^*DEF^/G^* - H^* +$

→ $ABC^*DEF^/G^* - H^* +$

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Infix to Prefix:

1. $A + (B * (C + (D * E)))$

Prefix:

- DE^*
 - CDE^*+
 - $BCDE^*+^*$
 - $ABCDE^*+^*+$
- $+A^*B+C*DE$

2. $A^*(B+C*(D-E)+F/G)$

Prefix:

- $DE-$
- $CDE-^*$
- $BCDE-^*+$
- $ABCDE-^*+^*$
- $FG/$
- $ABCDE-^*+^*FG/+$

Evaluate expressions:

1. $56 + 2 * 34 + -$

- $56 + \rightarrow 11$
- $112^* \rightarrow 22$
- $34 + \rightarrow 7$
- $227 - \rightarrow 15$

Result = 15

2. Prefix: $-^*45+23$

- $*45 \rightarrow 20$
- $+23 \rightarrow 5$
- $20-5 \rightarrow 15$

Result = 15

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3. Prefix: $+^*234$

$$\cdot +(*23)4 \rightarrow (2*3)+4 \rightarrow 6+4 = 10$$

result: 10

4. Postfix: $93^*42/$

$$\cdot 93^* \rightarrow 27$$

$$\cdot 42/ \rightarrow 2$$

$$\cdot 27-2 \rightarrow 25$$

result: 25